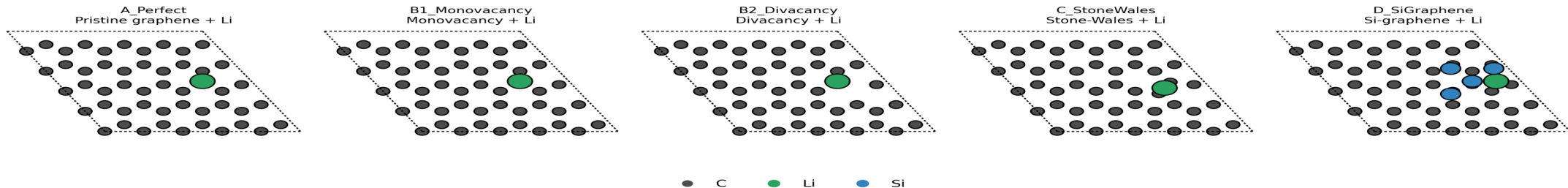


Validation-first AI for local lithium energetics

DFT anchors | leakage-aware MACE evaluation | out-of-domain stress tests



1 First-principles anchors

273

spin-polarized VASP frames

PBE-D3/dipole adsorption

-3.349 to 0.009 eV/Li

Relaxations + site/path scans + snapshot DFT

2 Audit before scale-up



Group-held-out split | E0 calibration | 3 seeds

3 Gate physical claims

SUPPORTED

Local adsorption screening

Model-domain diagnosis

WITHHELD

Unconverged CI-NEB barriers

Diffusion coefficients

9 DFT snapshot tests -> next training-data targets